CDA 106
Numerical Methods for Data Science

Dr Prashant Kumar Srivastava

IIT Patna

pksri@iitp.ac.in

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Solution of system of linear Algebraic Eq^ns

Drawback

If matrix 'A' is large, computations becomes time consuming.

- using method inverse $[AX = b \Rightarrow X = A^{-1}b]$
- Cramer's rule

→ Gauss Elimination method.

- Last class we solved some examples
- I'll provide an algorithm

Algorithm to solve a system using Gauss Elimination Method :-

$$A = [a_{ij}]$$

Step 1: Identify a 'pivot element' in the first column. [largest element in the column]

Step 2: Bring the pivot element to a_{11} position by using row exchange operations

Step 3: Make all the elements below the pivot element in the same column as ZERO, by using row operations.

Step 4 Now, repeat the same ~~of~~ steps in the second column onward without making any changes in the row of previous 'pivot element'

This will lead to the reduction of coefficient matrix to an upper triangular matrix.

Step 5 Apply "backward substitution" to get the values of unknown.

Example To solve using Gauss Elimination.

$$\underbrace{\begin{bmatrix} 2 & 1 & 1 & -2 \\ 4 & 0 & 2 & 1 \\ 3 & 2 & 2 & 0 \\ 1 & 3 & 2 & -1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} -10 \\ 8 \\ 7 \\ -5 \end{bmatrix}}_{b}$$

Ans

pivot

$$\left[\begin{array}{cccc|c} 2 & 1 & 1 & -2 & -10 \\ 4 & 0 & 2 & 1 & 8 \\ 3 & 2 & 2 & 0 & 7 \\ 1 & 3 & 2 & -1 & -5 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 8 \\ 2 & 1 & 1 & -2 & -10 \\ 3 & 2 & 2 & 0 & 7 \\ 1 & 3 & 2 & -1 & -5 \end{array}\right]$$

$$\xrightarrow{\substack{R_2 - \frac{1}{2}R_1 \\ R_3 - \frac{3}{4}R_1}} \quad R_4 - \frac{1}{4}R_1$$

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 8 \\ 0 & 1 & 0 & -5/2 & -14 \\ 0 & 2 & 1/2 & -3/4 & 1 \\ 0 & 3 & 3/2 & -5/4 & -7 \end{array}\right]$$

pivot

$$\xrightarrow{R_2 \leftrightarrow R_4} \left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 8 \\ 0 & 3 & 3/2 & -5/4 & -7 \\ 0 & 2 & 1/2 & -3/4 & 1 \\ 0 & 1 & 0 & -5/2 & -14 \end{array}\right]$$

$$\xrightarrow{\substack{R_3 - \frac{2}{3}R_2 \\ R_4 - \frac{1}{3}R_2}}$$

$$\left[\begin{array}{cccc|c} 4 & 6 & 2 & 1 & 8 \\ 0 & 3 & 3/2 & -5/4 & -7 \\ 0 & 0 & -1/2 & 1/12 & 17/3 \\ 0 & 0 & -1/2 & -\frac{25}{12} & -\frac{35}{3} \end{array}\right]$$

pivot

$$\xrightarrow{R_4 - R_3} \left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 8 \\ 0 & 3 & 3/2 & -5/4 & -7 \\ 0 & 0 & -1/2 & 1/12 & 17/3 \\ 0 & 0 & 0 & -13/6 & -\frac{52}{3} \end{array}\right]$$

Coeff. matrix is reduced to an upper triangular matrix.

Ans:
$x_1 = 5,\ x_2 = 6$
$x_3 = -10,\ x_4 = 8$

Backward Substitution

$$-\frac{13}{6} x_4 = -\frac{52}{3} \Rightarrow \boxed{x_4 = 8}$$

Next $\quad -\frac{1}{2} x_3 + \frac{1}{12} x_4 = \frac{17}{3}$

$$-\frac{1}{2} x_3 = \frac{17}{3} - \frac{8}{12} = \frac{15}{3} \Rightarrow \boxed{x_3 = -10}$$

$$4x_1 + 2x_3 + x_4 = 8$$

$$x_1 = \frac{1}{4}\left(8 - 2\times(-10) - 8\right) = 5$$

$$3x_2 + \frac{3}{2}x_3 - \frac{5}{4}x_4 = -7$$

$$x_2 = \frac{1}{3}\left(-7 - \frac{3}{2}\times(-10) + \frac{5}{4}(8)\right) = 6$$

In the calculation of Gauss Elimination method, you may have noticed that although the method is very simple and systematic, it is still complex (computationally involved)

Q: Can there be any method/way to deal with it?

Ans Yes! IDEA → Matrix Decomposition
or
Matrix Factorization

→ This matrix decomposition can be used to solve the system of linear algebraic eqn efficiently.

Let us recall

$U = \begin{bmatrix} \ddots & \\ 0 & \ddots \end{bmatrix} = [u_{ij}]$

upper triangular matrices

$u_{ij} = 0$ for $i > j$

$L = \begin{bmatrix} \ddots & 0 \\ & \ddots \end{bmatrix} = [l_{ij}]$

lower triangular matrices

$l_{ij} = 0$ for $i < j$

$D = \begin{bmatrix} \ddots & 0 \\ 0 & \ddots \end{bmatrix} = [d_{ij}]$

Diagonal matrix

$d_{ij} = 0 \quad i \neq j$

Consider $\underset{n\times n}{A}\ \underbrace{\underset{n\times 1}{X}}_{\text{unknown}} = \underset{n\times 1}{b}$

(Aim is to solve $AX = b$ or to find X)

$$A_{n\times n} \xrightarrow[\text{we factorize}]{\text{somehow}} L_{n\times n}\ U_{n\times n}$$

$$\underbrace{A_{n\times n}}\ X_{n\times 1} = b_{n\times 1}$$

$$L_{n\times n}\ \boxed{\underbrace{U_{n\times n}\ X_{n\times 1}}_{Y_{n\times 1}}} = b_{n\times 1}$$

$$\underline{L_{n\times n}}\ \underbrace{Y_{n\times 1}}_{\text{unknown}} = \underline{b_{n\times 1}}$$

forward substitution be applied

$Y_{n\times 1}$ known

$$\underbrace{U_{n\times n}\ \overset{\text{unknown}}{X_{n\times 1}} = \overset{\text{known}}{Y_{n\times 1}}}$$

Backward substitution

$X_{n\times 1}$ known

This is what you were looking for

Example

Given the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}$, verify that

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & -1 & 0 \\ 3 & 2 & -10 \end{bmatrix} \quad \& \quad U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

are

LU decomposition or LU factorization.

Need to show $A = LU$ (left as an exercise).

Solve

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix}}_{b}$$

using LU factorization

Ans

Since we know $A = LU$ where $L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & -1 & 0 \\ 3 & 2 & -10 \end{bmatrix}$ & $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$

[From previous example sl. 9]

Therefore, system $\underset{3\times3}{A}\,\underset{3\times1}{X} = \underset{3\times1}{b}$ gives

$$LUX = b$$

$$\Rightarrow LY = b \quad \text{where } UX = Y_{3\times1} = (y_1, y_2, y_3)^T$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 4 & -1 & 0 \\ 3 & 2 & -10 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix}$$

Now use forward substitution

$y_1 = 1$

$4y_1 - y_2 = 6 \Rightarrow y_2 = -2$

$3y_1 + 2y_2 - 10y_3 = 4 \Rightarrow y_3 = -1/2$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -1/2 \end{bmatrix}$$

Backward substitution

$\boxed{x_3 = -1/2}$

$x_2 + 5x_3 = -2 \Rightarrow \boxed{x_2 = 1/2}$

$x_1 + x_2 + x_3 = 1 \Rightarrow \boxed{x_1 = 1}$

Ans

$x_1 = 1$

$x_2 = 1/2$

$x_3 = -1/2$

Now, I am going to address the idea of how to factorize a matrix A into L & U matrices ?

<u>i.e.</u> $A = LU$!!

Consider a given matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Assume that

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \quad \& \quad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

If $A = LU$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ & \ldots & \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ \underline{\qquad\qquad} & & \\ \underline{\qquad\qquad} & & \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ \underline{\qquad\qquad} & & \\ \underline{\qquad\qquad} & & \end{bmatrix}$$

or $\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}}_{\text{known}} = \underbrace{\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}}_{\text{unknown}} \underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}}_{\text{unknown}}$ 12 unknown

From here, by taking simple product of RHS (L & U matrices) and comparing with LHS we get

① $a_{11} = l_{11} u_{11}$

② $a_{12} = l_{11} u_{12}$

③ $a_{13} = l_{11} u_{13}$

④ $a_{21} = l_{21} u_{11}$

⑤ $a_{22} = l_{21} u_{12} + l_{22} u_{22}$

⑥ $a_{23} = l_{21} u_{13} + l_{22} u_{23}$

⑦ $a_{31} = l_{31} u_{11}$

⑧ $a_{32} = l_{31} u_{12} + l_{32} u_{22}$

⑨ $a_{33} = l_{31} u_{13} + l_{32} u_{23} + l_{33} u_{33}$

- Here, l_{ij} & u_{ij} are unknown and a_{ij} are known
- Here, we have 9 eqns in 12 unknowns

Therefore, in order to get solution we can fix $12 - 9 = 3$ unknowns

We can solve above eq^n ①-⑨, by fixing.

(A) $l_{11} = l_{22} = l_{33} = 1$ [Do little LU decomposition]

$u_{11} = a_{11}$, $u_{12} = a_{12}$, $u_{13} = a_{13}$, $l_{21} = \frac{a_{21}}{u_{11}}$, $l_{31} = \frac{a_{31}}{u_{11}}$ ✓

$u_{22} = \frac{a_{22} - l_{21} u_{12}}{l_{22}}$, $u_{23} = \frac{a_{23} - l_{21} u_{13}}{l_{22}}$ ✓

$l_{32} = \frac{a_{32} - l_{31} u_{13}}{u_{22}}$, $u_{33} = \frac{a_{33} - l_{31} u_{13} - l_{32} u_{23}}{l_{33}}$ ✓

(B) $u_{11} = u_{22} = u_{33} = 1$ [Crout's LU decomposition]

Similarly as in case (A) we can find l_{ij} & u_{jj} in case B.